

Press Line — Operation & Reference Manual

Equipment Manual · Machine Type: Press Line

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Owner	Manufacturing Engineering
Applies To	Press Line — Plant A (Pune), Plant C (Ahmedabad); units Press_Line_1 through Press_Line_20

1. Overview

Hydraulic stamping presses used for sheet-metal forming operations. Presses on this line are the single largest source of major downtime plant-wide, predominantly due to hydraulic system failures (error code DT-01).

2. Machine Specifications

Parameter	Specification
Rated tonnage	150 – 400 tons
Stroke rate	8 – 20 strokes/min
Hydraulic system pressure	180 – 210 bar nominal
Ideal cycle time	1.0 – 1.35 min/part
Die clamping	Hydraulic quick-clamp, 4-point

3. Startup Procedure

- 1 Inspect hydraulic reservoir sight-glass for oil level; top up only with approved ISO VG 46 hydraulic oil.
- 2 Energize hydraulic pump and allow 3–5 minutes warm-up to reach operating temperature (35–45°C).
- 3 Verify safety light curtain and two-hand anti-tie-down control before first stroke.
- 4 Run 3 dry strokes with die open to confirm ram travel and pressure stability.

4. Hydraulic Health Monitoring

DT-01 (Hydraulic System Failure) accounts for the highest average downtime duration of any code recorded on this line — historically averaging over 2 hours per event when a seal or hose failure is left undetected. Operators must log pressure gauge readings every 2 hours per the Hydraulic_Maintenance_SOP.

5. Common Downtime Codes on This Machine Type

Code	Description	Severity	Reference SOP
DT-01	Hydraulic System Failure	Major	Hydraulic_Maintenance_SOP
DT-02	Raw Material / Feed Shortage	Major	PM_Guide (logistics escalation)
DT-03	Tooling / Die Changeover Delay	Moderate	Changeover_SOP
DT-05	Conveyor Jam / Misfeed	Minor	Inspection_Checklist

■ **Never place hands or tools in the die area during pump-on state. Verify zero-energy state before any hydraulic line disconnection.**